

THE NEW ALBA DIFFRACTOMETER FOR MICROFOCUS BEAM MACROMOLECULAR CRYSTALLOGRAPHY EXPERIMENTS AT XAIRA BEAMLINE

N. Gonzalez, D. Garriga, I. Crespo, J. L. Frieiro, N. Pereira, B. Molás, J. Gabadinho, I. Sics, A. Carballedo, C. Colldelram, J. Nicolás, J. Juanhuix
ALBA Synchrotron Light Source, Cerdanyola del Valles, Spain

Abstract

XAIRA, the new microfocus MX beamline at ALBA aims to deliver optimal diffraction images by enclosing the entire end-station in He atmosphere, including the diffractometer and the detector, while keeping the compatibility with standard cryo-crystallography tools and robot. The sub-100 nm SoC diffractometer, based on a unique helium bearing goniometer also compatible with air, has been designed to deliver high quality data from micron-sized crystals from fast oscillation and fixed-target MX experiments while allowing a tight sample-to-detector distance of 70mm. The diffractometer also includes a double on-axis visualization system for sample imaging at sub-micron resolution, a quick retractable collimator and beamstop assembly, a front- and backlight illumination system and a fast in/out YAG:Ce screen system for beam positioning. Here, the overall system design and performance results are presented.

INTRODUCTION

The new BL06-XAIRA microfocus macromolecular crystallography (MX) beamline at the ALBA synchrotron light facility, which has entered into user operation this year, provides a 3.5-14 keV stable, high flux beam which could be focused down to $3 \times 1 \mu\text{m}^2$ [1]. The beamline supports single-axis rotation experiments, including helical scans, and fixed-target serial crystallography (SSX).

A key feature of the beamline is a chamber enclosing the whole end station, which allows experiments to be performed either in air or in helium atmosphere, and both at room temperature or under cryogenic conditions. The helium environment not only enables experiments in the low energy range below 6 keV, thus providing optimal conditions for native anomalous experiments, but also benefits the experiments in the whole energy range by eliminating the background from air scattering.

This end station chamber includes the sample environment, diffractometer, cryostream and detector, and it is compatible with automated sample mounting and the standard cryo crystallography sample holders. The components of the diffractometer, an omega axis goniometer and the sample conditioning and monitoring components, are supported over a stable omega XY stage adjusting the whole setup together, see Fig. 1.

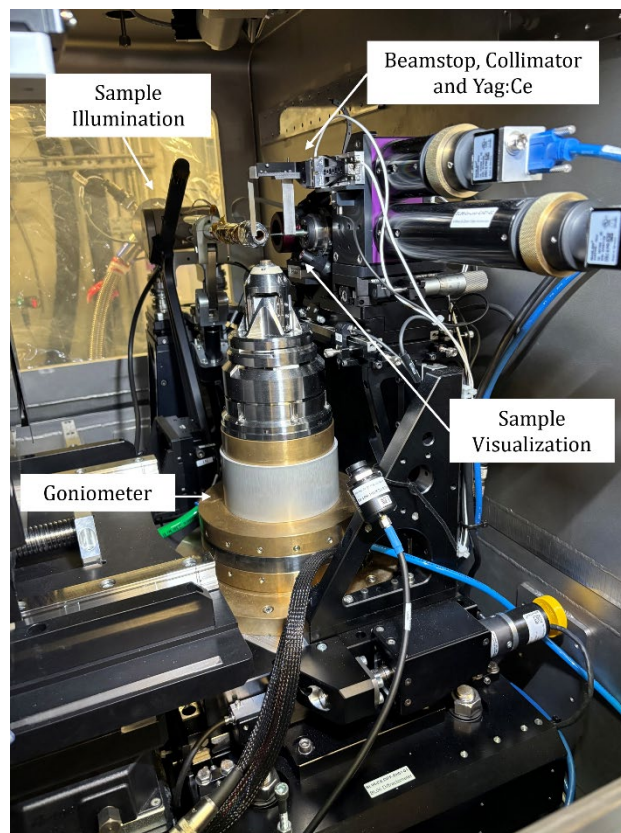


Figure 1: Diffractometer components within the helium chamber.

SYSTEM DESCRIPTION

Goniometer

The central component of the diffractometer is a low friction vertical single-axis goniometer, designed at ALBA to comply with the tight geometrical requirements of the end station and manufactured by Fluid Film Devices Ltd. The goniometer bearing can be fed either by air or He with similar performance. The goniometer is actuated by a direct drive slotless torque motor (Aerotech S-130), which has no cogging torque. The position and speed control loops are closed with a double head ultra-high accuracy rotation analogue encoder (Renishaw REXM20), which meets the requirements in terms of maximum resolution, 0.04 mdeg, and maximum speed, 60 rpm. The goniometer is installed on an in-house designed, 0.1 μm resolution and high stability XY stage, actuated by stepper motors. This stage permits positioning the goniometer axis at the focal

point of the beam or moving it along the beam path to have a defocused beam at sample position. On top of the goniometer, a fast high resolution XYZ setup based on piezo stages (SMARACT SLC Series) with 10nm resolution and mounted on a titanium frame to minimize the thermal drifts, allows not only centering the sample with respect to the rotation axis but also performing any trajectory (e.g. helical) to be used in rotation experiments. The omega axis goniometer and the piezo stages are controlled using a master motion controller (Aerotech A3200), see Fig. 2.

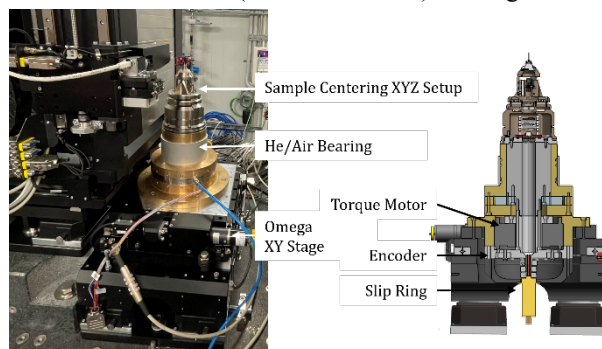


Figure 2: Goniometer subsystem components.

Additionally, the centering stages setup can be easily removed so as to install a dedicated fast sample rastering XZ stage to optimally perform fixed-target SSX experiments from large chips at 10mm/s speed with 10 nm resolution. The SSX is currently being tested and it will be ready to be installed at the beamline by the end of the year.

Sample Visualization System

The sample visualization setup of XAIRA is based on a double on-axis visualization (OAV) system. On the one hand, a commercial OAV system (BZoom, ARINAX) with a variable magnification and with a 1.4 mm diameter hole that allows coarse and fine sample centering with respect to the beam. On the other hand, an undrilled, high-magnification microscope allows sample imaging at sub-micron resolution to visualize micron-sized crystals. Blue monochromatic illumination is being explored to further improve the spatial resolution by reducing the chromatic aberrations and the diffraction effects. The two subsystems are located on a high stability support over the base XY stage, so that they move together with the goniometer and the sample. This support includes a motorized Z stage to adjust the visualization system position vertically. In addition, a fast X stage, transversal to the beam, permits a fast exchange from one system to the other in less than 3 s. The distance in the beam direction between the commercial OAV system and the goniometer axis is adjusted manually during installation with a high-resolution stage whereas the position of the high-resolution microscope can be remotely adjusted with a piezo stage.

Beamstop, Collimator and YAG:Ce Screen

Mounted on top of the commercial OAV system, two XZ piezo stages permit the accurate positioning of the custom designed beamstop and collimator assembly. Furthermore, the X stage has a range long enough to move the whole assembly out fast for the sample loading coming from the top. The design of the beamstop and collimator ensures the compatibility with sample chips of up to 40x40 mm² and the SSX setup. On the sample support, a rotary piezo stage allows locating a small YAG:Ce screen at sample position which serves as a fast beam positioning method. An additional Y piezo stage allows the positioning of the screen at the same position of the sample (Fig. 3).

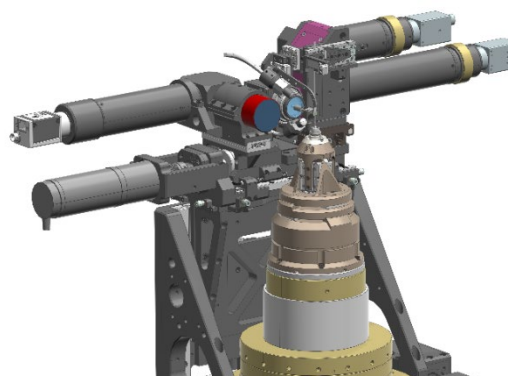


Figure 3: Image of the beamstop, collimator and Yag:Ce assembly, mounted over the double sample visualization system.

Sample Illumination

As for the sample illumination, two commercial components from ARINAX have been integrated in the diffractometer: a front light illumination system, which is mounted on the OAV system, and a backlight, which is located over an in-out stage over the goniometer support base. Due to the larger spot size of the backlight, it is not required to be moved together with the XY stage of the goniometer, so it's stage is mounted out so as to also improve the stability of the whole system.

GONIOMETER RUN-OUT METROLOGY AND CORRECTION

The small beam size at sample position, $3 \times 1 \mu\text{m}^2$, and the micron-sized dimensions of the crystals to be analyzed at XAIRA impose a strict requirement on the radial runout of the goniometer, which is actually one of the most critical performance parameters of the beamline.

In order to measure the runout, a new method has been developed at ALBA, based on a computer vision scheme. Using a video-microscope looking to the rotation plane and post processing the acquired images, the radial run out of the goniometer is obtained, with nanometer resolution [2].

The first measurements, taken for different working speeds from 5 to 360 deg/s showed a runout above specified values (± 50 nm). The main source of radial runout was actually the slip ring, which added around 100 nm, but also, the 18 poles of the motor rotor were also visible in the plots, as shown in Fig. 4.

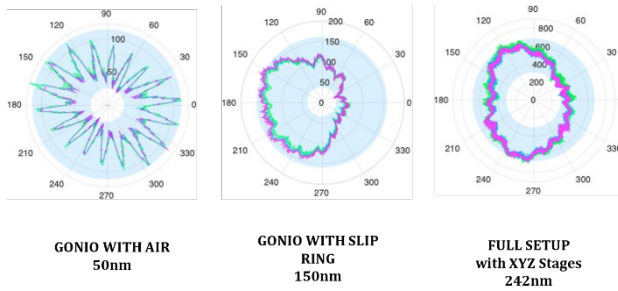


Figure 4: Run out metrology results at 360 deg/s, with just the goniometer (left), goniometer with slirping (middle) and full setup, including the XYZ stages on top (right).

To reduce the runout and comply with the specifications, a low-level feed-forward correction of the systematic errors of the goniometer using the XY piezo stages was implemented. Based on the previously taken measurements, a series of lookup tables at different goniometer speeds and for air and helium flows was implemented (Fig. 5).

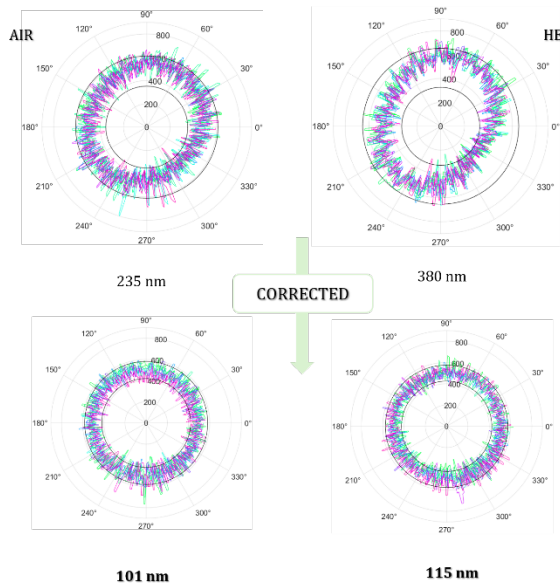


Figure 5: Run out metrology plots at 360 deg/s, with (down) and without (top) correction and with the bearing fed with air (left) and with helium (right).

The goniometer is designed to have the main axis decoupled from the slip ring, which could be run with a slave motor to mitigate the perturbations on the sample. This option has been not implemented so far, as the corrected values of the runout align with the required specs.

The measured runout at different angular speeds is shown in Tables 1 and 2 for air and helium flows, respectively. The values include the background noise of the measurement, estimated to be ± 15 nm for static measurements. The uncorrected runout at 5 deg/s was not properly

measured, so its correction is calculated from the 10 deg/s measurement.

Table 1: Total Peak-to-Valley Runout Values with and without Correction for Air, in nm.

Speed [deg/s]	Without Correction	After Correction
5	N/A	± 36.41
10	± 105.34	± 28.73
40	± 129.84	± 30.46
80	± 134.57	± 28.89
120	± 112.63	± 41.50
200	± 122.47	± 41.53
280	± 119.74	± 33.30
360	± 117.58	± 50.75

Table 2: Total Peak-to-Valley Runout Values with and without Correction for Helium, in nm.

Speed [deg/s]	Without Correction	After Correction
5	N/A	± 54.68
10	± 232.22	± 35.22
40	± 172.20	± 67.73
80	± 156.80	± 60.35
120	± 163.62	± 45.60
200	± 167.21	± 45.50
280	± 185.22	± 75.01
360	± 190.31	± 57.72

CONCLUSIONS

The design of the ALBA diffractometer, installed at the microfocus MX Beamline XAIRA Beamline, has been described, including the runout measurement results that validate the performance of the system in helium and air. In parallel with the ongoing user operation, further commissioning for optimization and the implementation of the fixed-target SSX setup will be performed during this year.

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